

ЛР7

План запроса 1 до:

"QUERY PLAN"

"Merge Left Join (cost=9635.90..11083.65 rows=100 width=126) (actual time=145.134..155.691 rows=100 loops=1)"

" Merge Cond: (b.bakery_id = bis.bakery_id)"

" Buffers: shared hit=5808 read=52"

" -> Merge Left Join (cost=9635.62..10965.37 rows=100 width=118) (actual time=145.097..154.949 rows=100 loops=1)"

" Merge Cond: (b.bakery_id = co.bakery_id)"

" Buffers: shared hit=5788 read=52"

" -> Sort (cost=7.32..7.57 rows=100 width=70) (actual time=0.059..0.068 rows=100 loops=1)"

" Sort Key: b.bakery_id"

" Sort Method: quicksort Memory: 33kB"

" Buffers: shared hit=3"

" -> Seq Scan on bakery b (cost=0.00..4.00 rows=100 width=70) (actual time=0.018..0.037 rows=100 loops=1)"

" Buffers: shared hit=3"

" -> GroupAggregate (cost=9628.30..10955.29 rows=100 width=56) (actual time=145.035..154.816 rows=100 loops=1)"

" Group Key: co.bakery_id"

" Buffers: shared hit=5785 read=52"

" -> Gather Merge (cost=9628.30..10849.21 rows=10483 width=30) (actual time=144.982..151.752 rows=9684 loops=1)"

" Workers Planned: 2"

" Workers Launched: 2"

" Buffers: shared hit=5785 read=52"

" -> Sort (cost=8628.27..8639.19 rows=4368 width=30) (actual time=33.042..33.180 rows=3228 loops=3)"

" Sort Key: co.bakery_id, co.order_id"

" Sort Method: quicksort Memory: 544kB"

" Buffers: shared hit=5785 read=52"

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"           Worker 0: Sort Method: quicksort Memory: 199kB"
"           Worker 1: Sort Method: quicksort Memory: 25kB"
"           -> Parallel Seq Scan on customer_order co (cost=0.00..8364.17 rows=4368
width=30) (actual time=0.020..31.565 rows=3228 loops=3)"
"           Filter: (status = ANY ('{COMPLETED,DELIVERED}'::text[]))"
"           Rows Removed by Filter: 163439"
"           Buffers: shared hit=5711 read=52"
" -> GroupAggregate (cost=0.28..115.78 rows=100 width=16) (actual time=0.035..0.693
rows=100 loops=1)"
"   Group Key: bis.bakery_id"
"   Buffers: shared hit=20"
"   -> Index Only Scan using pk_bakery_ingredient_stock on bakery_ingredient_stock bis
(cost=0.28..102.78 rows=2400 width=16) (actual time=0.022..0.421 rows=2400 loops=1)"
"     Heap Fetches: 1080"
"     Buffers: shared hit=20"
"Planning Time: 0.262 ms"
"Execution Time: 155.871 ms"

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План запроса 5 до:

"QUERY PLAN"

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"WindowAgg (cost=39663.96..39664.19 rows=7 width=188) (actual time=262.927..266.423
rows=7 loops=1)"
" Buffers: shared hit=7532 read=14018"
" -> WindowAgg (cost=39663.93..39664.05 rows=7 width=188) (actual
time=262.921..266.413 rows=7 loops=1)"
"   Buffers: shared hit=7532 read=14018"
"   -> Sort (cost=39663.91..39663.93 rows=7 width=156) (actual time=262.909..266.394
rows=7 loops=1)"
"     Sort Key: (COALESCE((sum((sum(oi.line_total_amount)))), '0'::numeric)) DESC,
p.name"
"     Sort Method: quicksort Memory: 25kB"
"     Buffers: shared hit=7532 read=14018"
"     -> Merge Left Join (cost=34717.61..39663.81 rows=7 width=156) (actual
time=229.271..266.375 rows=7 loops=1)"
"       Merge Cond: (p.name = p_4.name)"

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"          Buffers: shared hit=7532 read=14018"
"          -> Merge Left Join (cost=34632.77..39565.63 rows=7 width=116) (actual
time=227.879..264.667 rows=7 loops=1)"
"          Merge Cond: (p.name = p_3.name)"
"          Buffers: shared hit=7510 read=14018"
"          -> Merge Left Join (cost=840.20..906.25 rows=7 width=36) (actual
time=9.772..11.463 rows=7 loops=1)"
"          Merge Cond: (p.name = p_2.name)"
"          Buffers: shared hit=130"
"          -> Merge Left Join (cost=518.07..557.80 rows=7 width=28) (actual
time=6.859..7.896 rows=7 loops=1)"
"          Merge Cond: (p.name = p_1.name)"
"          Buffers: shared hit=72"
"          -> Sort (cost=20.92..20.94 rows=7 width=20) (actual
time=0.386..0.390 rows=7 loops=1)"
"          Sort Key: p.name"
"          Sort Method: quicksort Memory: 25kB"
"          Buffers: shared hit=12"
"          -> HashAggregate (cost=20.75..20.82 rows=7 width=20) (actual
time=0.363..0.365 rows=7 loops=1)"
"          Group Key: p.name"
"          Batches: 1 Memory Usage: 24kB"
"          Buffers: shared hit=12"
"          -> Seq Scan on product p (cost=0.00..19.00 rows=700
width=20) (actual time=0.019..0.103 rows=700 loops=1)"
"          Buffers: shared hit=12"
"          -> GroupAggregate (cost=497.15..536.69 rows=7 width=28) (actual
time=6.470..7.481 rows=7 loops=1)"
"          Group Key: p_1.name"
"          Buffers: shared hit=60"
"          -> Sort (cost=497.15..510.31 rows=5262 width=28) (actual
time=6.353..6.660 rows=5400 loops=1)"
"          Sort Key: p_1.name, plii.ingredient_id"
"          Sort Method: quicksort Memory: 447kB"
"          Buffers: shared hit=60"

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"          -> Hash Join (cost=51.50..171.92 rows=5262 width=28) (actual
time=0.683..3.028 rows=5400 loops=1)"
"
"          Hash Cond: (pli.product_id = p_1.product_id)"
"
"          Buffers: shared hit=60"
"
"          -> Hash Join (cost=23.75..130.27 rows=5262 width=16)
(actual time=0.268..1.692 rows=5400 loops=1)"
"
"          Hash Cond: (plii.price_list_item_id =
pli.price_list_item_id)"
"
"          Buffers: shared hit=48"
"
"          -> Seq Scan on price_list_item_ingredient plii
(cost=0.00..92.62 rows=5262 width=16) (actual time=0.009..0.308 rows=5400 loops=1)"
"
"          Buffers: shared hit=40"
"
"          -> Hash (cost=15.00..15.00 rows=700 width=16)
(actual time=0.252..0.253 rows=700 loops=1)"
"
"          Buckets: 1024 Batches: 1 Memory Usage: 41kB"
"
"          Buffers: shared hit=8"
"
"          -> Seq Scan on price_list_item pli (cost=0.00..15.00
rows=700 width=16) (actual time=0.009..0.122 rows=700 loops=1)"
"
"          Buffers: shared hit=8"
"
"          -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.345..0.346 rows=700 loops=1)"
"
"          Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"
"          Buffers: shared hit=12"
"
"          -> Seq Scan on product p_1 (cost=0.00..19.00
rows=700 width=28) (actual time=0.021..0.144 rows=700 loops=1)"
"
"          Buffers: shared hit=12"
"
"          -> GroupAggregate (cost=322.13..348.27 rows=7 width=28) (actual
time=2.911..3.548 rows=7 loops=1)"
"
"          Group Key: p_2.name"
"
"          Buffers: shared hit=58"
"
"          -> Sort (cost=322.13..330.82 rows=3476 width=28) (actual
time=2.841..3.035 rows=3500 loops=1)"
"
"          Sort Key: p_2.name, pu.utensil_id"
"
"          Sort Method: quicksort Memory: 261kB"
"
"          Buffers: shared hit=58"

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"                -> Hash Join (cost=27.75..117.69 rows=3476 width=28) (actual
time=0.166..0.939 rows=3500 loops=1)"
"
"                Hash Cond: (pu.product_id = p_2.product_id)"
"
"                Buffers: shared hit=58"
"
"                -> Seq Scan on product_utensil pu (cost=0.00..80.76
rows=3476 width=16) (actual time=0.019..0.186 rows=3500 loops=1)"
"
"                Buffers: shared hit=46"
"
"                -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.140..0.141 rows=700 loops=1)"
"
"                Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"
"                Buffers: shared hit=12"
"
"                -> Seq Scan on product p_2 (cost=0.00..19.00 rows=700
width=28) (actual time=0.007..0.071 rows=700 loops=1)"
"
"                Buffers: shared hit=12"
"
"                -> GroupAggregate (cost=33792.56..38656.79 rows=200 width=100) (actual
time=218.105..253.169 rows=7 loops=1)"
"
"                Group Key: p_3.name"
"
"                Buffers: shared hit=7380 read=14018"
"
"                -> Finalize GroupAggregate (cost=33792.56..37866.87 rows=31457
width=76) (actual time=213.536..245.710 rows=24112 loops=1)"
"
"                Group Key: p_3.name, co.order_id"
"
"                Buffers: shared hit=7380 read=14018"
"
"                -> Gather Merge (cost=33792.56..37145.98 rows=26214 width=76)
(actual time=213.525..230.018 rows=24121 loops=1)"
"
"                Workers Planned: 2"
"
"                Workers Launched: 2"
"
"                Buffers: shared hit=7380 read=14018"
"
"                -> Partial GroupAggregate (cost=32792.54..33120.21 rows=13107
width=76) (actual time=176.497..182.860 rows=8040 loops=3)"
"
"                Group Key: p_3.name, co.order_id"
"
"                Buffers: shared hit=7380 read=14018"
"
"                -> Sort (cost=32792.54..32825.30 rows=13107 width=46)
(actual time=176.484..177.288 rows=9692 loops=3)"
"
"                Sort Key: p_3.name, co.order_id"
"
"                Sort Method: quicksort Memory: 1064kB"

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" Buffers: shared hit=7380 read=14018"

" Worker 0: Sort Method: quicksort Memory: 1073kB"

" Worker 1: Sort Method: quicksort Memory: 1062kB"

" -> Hash Join (cost=8470.27..31896.15 rows=13107
width=46) (actual time=30.646..159.020 rows=9692 loops=3)"

" Hash Cond: (pli_1.product_id = p_3.product_id)"

" Buffers: shared hit=7290 read=14018"

" -> Hash Join (cost=8442.52..31833.78 rows=13107
width=34) (actual time=29.984..156.061 rows=9692 loops=3)"

" Hash Cond: (oi.price_list_item_id =
pli_1.price_list_item_id)"

" Buffers: shared hit=7230 read=14018"

" -> Parallel Hash Join (cost=8418.77..31775.41
rows=13107 width=34) (actual time=29.458..153.146 rows=9692 loops=3)"

" Hash Cond: (oi.order_id = co.order_id)"

" Buffers: shared hit=7206 read=14018"

" -> Parallel Seq Scan on order_item oi
(cost=0.00..21715.59 rows=625159 width=26) (actual time=0.663..59.211 rows=500127
loops=3)"

" Buffers: shared hit=1446 read=14018"

" -> Parallel Hash (cost=8364.17..8364.17
rows=4368 width=16) (actual time=28.597..28.598 rows=3228 loops=3)"

" Buckets: 16384 Batches: 1 Memory Usage:
608kB"

" Buffers: shared hit=5760"

" -> Parallel Seq Scan on customer_order co
(cost=0.00..8364.17 rows=4368 width=16) (actual time=0.020..27.519 rows=3228 loops=3)"

" Filter: (status = ANY
('{COMPLETED,DELIVERED}'::text[]))"

" Rows Removed by Filter: 163439"

" Buffers: shared hit=5760"

" -> Hash (cost=15.00..15.00 rows=700 width=16)
(actual time=0.513..0.514 rows=700 loops=3)"

" Buckets: 1024 Batches: 1 Memory Usage:
41kB"

" Buffers: shared hit=24"

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"                                -> Seq Scan on price_list_item pli_1
(cost=0.00..15.00 rows=700 width=16) (actual time=0.316..0.410 rows=700 loops=3)"
"                                Buffers: shared hit=24"
"                                -> Hash (cost=19.00..19.00 rows=700 width=28)
(actual time=0.629..0.629 rows=700 loops=3)"
"                                Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"                                Buffers: shared hit=36"
"                                -> Seq Scan on product p_3 (cost=0.00..19.00
rows=700 width=28) (actual time=0.433..0.522 rows=700 loops=3)"
"                                Buffers: shared hit=36"
"                                -> GroupAggregate (cost=84.84..95.59 rows=200 width=28) (actual
time=1.388..1.666 rows=7 loops=1)"
"                                Group Key: p_4.name"
"                                Buffers: shared hit=22"
"                                -> Unique (cost=84.84..90.09 rows=700 width=28) (actual time=1.354..1.519
rows=700 loops=1)"
"                                Buffers: shared hit=22"
"                                -> Sort (cost=84.84..86.59 rows=700 width=28) (actual time=1.353..1.420
rows=700 loops=1)"
"                                Sort Key: p_4.name, pl.bakery_id"
"                                Sort Method: quicksort Memory: 57kB"
"                                Buffers: shared hit=22"
"                                -> Hash Join (cost=33.00..51.76 rows=700 width=28) (actual
time=0.309..0.581 rows=700 loops=1)"
"                                Hash Cond: (pli_2.price_list_id = pl.price_list_id)"
"                                Buffers: shared hit=22"
"                                -> Hash Join (cost=27.75..44.60 rows=700 width=28) (actual
time=0.189..0.363 rows=700 loops=1)"
"                                Hash Cond: (pli_2.product_id = p_4.product_id)"
"                                Buffers: shared hit=20"
"                                -> Seq Scan on price_list_item pli_2 (cost=0.00..15.00
rows=700 width=16) (actual time=0.008..0.049 rows=700 loops=1)"
"                                Buffers: shared hit=8"
"                                -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.175..0.177 rows=700 loops=1)"

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"                                Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"                                Buffers: shared hit=12"
"                                -> Seq Scan on product p_4 (cost=0.00..19.00 rows=700
width=28) (actual time=0.008..0.090 rows=700 loops=1)"
"                                Filter: is_active"
"                                Buffers: shared hit=12"
"                                -> Hash (cost=4.00..4.00 rows=100 width=16) (actual
time=0.052..0.052 rows=100 loops=1)"
"                                Buckets: 1024 Batches: 1 Memory Usage: 13kB"
"                                Buffers: shared hit=2"
"                                -> Seq Scan on price_list pl (cost=0.00..4.00 rows=100
width=16) (actual time=0.024..0.038 rows=100 loops=1)"
"                                Filter: (is_active AND (valid_from <= CURRENT_DATE)
AND ((valid_to IS NULL) OR (valid_to >= CURRENT_DATE)))"
"                                Buffers: shared hit=2"
"Planning:"
" Buffers: shared hit=60"
"Planning Time: 2.207 ms"
"Execution Time: 267.245 ms"

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План запроса 1 после индексов:

"QUERY PLAN"

"Merge Left Join (cost=1171.01..1424.05 rows=100 width=126) (actual time=5.579..8.942 rows=100 loops=1)"

" Merge Cond: (b.bakery_id = bis.bakery_id)"

" Buffers: shared hit=4856"

" -> Merge Left Join (cost=1170.73..1305.77 rows=100 width=118) (actual time=5.560..8.393 rows=100 loops=1)"

" Merge Cond: (b.bakery_id = co.bakery_id)"

" Buffers: shared hit=4836"

" -> Sort (cost=7.32..7.57 rows=100 width=70) (actual time=0.038..0.045 rows=100 loops=1)"

" Sort Key: b.bakery_id"

" Sort Method: quicksort Memory: 33kB"

" Buffers: shared hit=3"


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"      -> Seq Scan on bakery b (cost=0.00..4.00 rows=100 width=70) (actual
time=0.012..0.025 rows=100 loops=1)"
"      Buffers: shared hit=3"
"      -> GroupAggregate (cost=1163.41..1295.70 rows=100 width=56) (actual
time=5.519..8.314 rows=100 loops=1)"
"      Group Key: co.bakery_id"
"      Buffers: shared hit=4833"
"      -> Sort (cost=1163.41..1189.62 rows=10483 width=30) (actual time=5.477..5.784
rows=9684 loops=1)"
"      Sort Key: co.bakery_id, co.order_id"
"      Sort Method: quicksort Memory: 838kB"
"      Buffers: shared hit=4833"
"      -> Index Scan using idx_lab_co_done_order_cover on customer_order co
(cost=0.29..463.37 rows=10483 width=30) (actual time=0.011..3.972 rows=9684 loops=1)"
"      Buffers: shared hit=4833"
"      -> GroupAggregate (cost=0.28..115.78 rows=100 width=16) (actual time=0.017..0.518
rows=100 loops=1)"
"      Group Key: bis.bakery_id"
"      Buffers: shared hit=20"
"      -> Index Only Scan using pk_bakery_ingredient_stock on bakery_ingredient_stock bis
(cost=0.28..102.78 rows=2400 width=16) (actual time=0.012..0.293 rows=2400 loops=1)"
"      Heap Fetches: 1080"
"      Buffers: shared hit=20"
"Planning Time: 0.285 ms"
"Execution Time: 9.131 ms"

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План запроса 5 после индексов:

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"QUERY PLAN"
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```
"WindowAgg (cost=28113.54..28113.77 rows=7 width=188) (actual time=160.216..160.451
rows=7 loops=1)"
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" Buffers: shared hit=34700"
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```
" -> WindowAgg (cost=28113.51..28113.63 rows=7 width=188) (actual
time=160.210..160.441 rows=7 loops=1)"
```

```
" Buffers: shared hit=34700"
```

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"      -> Sort (cost=28113.49..28113.50 rows=7 width=156) (actual time=160.202..160.426
rows=7 loops=1)"
"          Sort Key: (COALESCE((sum((sum(oi.line_total_amount))))), '0'::numeric)) DESC,
p.name"
"          Sort Method: quicksort  Memory: 25kB"
"          Buffers: shared hit=34700"
"      -> Merge Left Join (cost=23167.19..28113.39 rows=7 width=156) (actual
time=121.132..160.410 rows=7 loops=1)"
"          Merge Cond: (p.name = p_4.name)"
"          Buffers: shared hit=34700"
"      -> Merge Left Join (cost=23082.34..28015.21 rows=7 width=116) (actual
time=119.713..158.745 rows=7 loops=1)"
"          Merge Cond: (p.name = p_3.name)"
"          Buffers: shared hit=34678"
"      -> Merge Left Join (cost=840.20..906.25 rows=7 width=36) (actual
time=9.015..10.410 rows=7 loops=1)"
"          Merge Cond: (p.name = p_2.name)"
"          Buffers: shared hit=130"
"      -> Merge Left Join (cost=518.07..557.80 rows=7 width=28) (actual
time=5.954..6.745 rows=7 loops=1)"
"          Merge Cond: (p.name = p_1.name)"
"          Buffers: shared hit=72"
"      -> Sort (cost=20.92..20.94 rows=7 width=20) (actual
time=0.198..0.202 rows=7 loops=1)"
"          Sort Key: p.name"
"          Sort Method: quicksort  Memory: 25kB"
"          Buffers: shared hit=12"
"      -> HashAggregate (cost=20.75..20.82 rows=7 width=20) (actual
time=0.180..0.181 rows=7 loops=1)"
"          Group Key: p.name"
"          Batches: 1  Memory Usage: 24kB"
"          Buffers: shared hit=12"
"      -> Seq Scan on product p (cost=0.00..19.00 rows=700
width=20) (actual time=0.013..0.054 rows=700 loops=1)"
"          Buffers: shared hit=12"

```

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"          -> GroupAggregate (cost=497.15..536.69 rows=7 width=28) (actual
time=5.753..6.527 rows=7 loops=1)"
"          Group Key: p_1.name"
"          Buffers: shared hit=60"
"          -> Sort (cost=497.15..510.31 rows=5262 width=28) (actual
time=5.635..5.896 rows=5400 loops=1)"
"          Sort Key: p_1.name, plii.ingredient_id"
"          Sort Method: quicksort  Memory: 447kB"
"          Buffers: shared hit=60"
"          -> Hash Join (cost=51.50..171.92 rows=5262 width=28) (actual
time=0.308..2.337 rows=5400 loops=1)"
"          Hash Cond: (pli.product_id = p_1.product_id)"
"          Buffers: shared hit=60"
"          -> Hash Join (cost=23.75..130.27 rows=5262 width=16)
(actual time=0.153..1.386 rows=5400 loops=1)"
"          Hash Cond: (plii.price_list_item_id =
pli.price_list_item_id)"
"          Buffers: shared hit=48"
"          -> Seq Scan on price_list_item_ingredient plii
(cost=0.00..92.62 rows=5262 width=16) (actual time=0.008..0.263 rows=5400 loops=1)"
"          Buffers: shared hit=40"
"          -> Hash (cost=15.00..15.00 rows=700 width=16)
(actual time=0.136..0.137 rows=700 loops=1)"
"          Buckets: 1024  Batches: 1  Memory Usage: 41kB"
"          Buffers: shared hit=8"
"          -> Seq Scan on price_list_item pli (cost=0.00..15.00
rows=700 width=16) (actual time=0.005..0.064 rows=700 loops=1)"
"          Buffers: shared hit=8"
"          -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.150..0.151 rows=700 loops=1)"
"          Buckets: 1024  Batches: 1  Memory Usage: 52kB"
"          Buffers: shared hit=12"
"          -> Seq Scan on product p_1 (cost=0.00..19.00
rows=700 width=28) (actual time=0.005..0.065 rows=700 loops=1)"
"          Buffers: shared hit=12"

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"          -> GroupAggregate (cost=322.13..348.27 rows=7 width=28) (actual
time=3.059..3.647 rows=7 loops=1)"
"          Group Key: p_2.name"
"          Buffers: shared hit=58"
"          -> Sort (cost=322.13..330.82 rows=3476 width=28) (actual
time=2.979..3.188 rows=3500 loops=1)"
"          Sort Key: p_2.name, pu.utensil_id"
"          Sort Method: quicksort Memory: 261kB"
"          Buffers: shared hit=58"
"          -> Hash Join (cost=27.75..117.69 rows=3476 width=28) (actual
time=0.172..0.980 rows=3500 loops=1)"
"          Hash Cond: (pu.product_id = p_2.product_id)"
"          Buffers: shared hit=58"
"          -> Seq Scan on product_utensil pu (cost=0.00..80.76
rows=3476 width=16) (actual time=0.021..0.199 rows=3500 loops=1)"
"          Buffers: shared hit=46"
"          -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.143..0.144 rows=700 loops=1)"
"          Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"          Buffers: shared hit=12"
"          -> Seq Scan on product p_2 (cost=0.00..19.00 rows=700
width=28) (actual time=0.007..0.073 rows=700 loops=1)"
"          Buffers: shared hit=12"
"          -> GroupAggregate (cost=22242.14..27106.37 rows=200 width=100) (actual
time=110.674..148.287 rows=7 loops=1)"
"          Group Key: p_3.name"
"          Buffers: shared hit=34548"
"          -> Finalize GroupAggregate (cost=22242.14..26316.45 rows=31457
width=76) (actual time=103.936..141.153 rows=24112 loops=1)"
"          Group Key: p_3.name, co.order_id"
"          Buffers: shared hit=34548"
"          -> Gather Merge (cost=22242.14..25595.56 rows=26214 width=76)
(actual time=103.929..125.746 rows=24112 loops=1)"
"          Workers Planned: 2"
"          Workers Launched: 2"

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"          Buffers: shared hit=34548"
"          -> Partial GroupAggregate (cost=21242.12..21569.79 rows=13107
width=76) (actual time=39.798..46.108 rows=8037 loops=3)"
"          Group Key: p_3.name, co.order_id"
"          Buffers: shared hit=34548"
"          -> Sort (cost=21242.12..21274.88 rows=13107 width=46)
(actual time=39.786..40.606 rows=9692 loops=3)"
"          Sort Key: p_3.name, co.order_id"
"          Sort Method: quicksort  Memory: 2692kB"
"          Buffers: shared hit=34548"
"          Worker 0: Sort Method: quicksort  Memory: 132kB"
"          Worker 1: Sort Method: quicksort  Memory: 62kB"
"          -> Hash Join (cost=263.25..20345.72 rows=13107
width=46) (actual time=2.301..21.921 rows=9692 loops=3)"
"          Hash Cond: (pli_1.product_id = p_3.product_id)"
"          Buffers: shared hit=34516"
"          -> Hash Join (cost=235.50..20283.36 rows=13107
width=34) (actual time=1.492..18.762 rows=9692 loops=3)"
"          Hash Cond: (oi.price_list_item_id =
pli_1.price_list_item_id)"
"          Buffers: shared hit=34456"
"          -> Nested Loop (cost=211.75..20224.99
rows=13107 width=34) (actual time=1.032..15.731 rows=9692 loops=3)"
"          Buffers: shared hit=34432"
"          -> Parallel Bitmap Heap Scan on customer_order
co (cost=211.33..6146.97 rows=4368 width=16) (actual time=0.931..3.582 rows=3228
loops=3)"
"          Recheck Cond: (status = ANY
('{COMPLETED,DELIVERED}'::text[]))"
"          Heap Blocks: exact=4460"
"          Buffers: shared hit=4769"
"          -> Bitmap Index Scan on
idx_lab_co_done_order_cover (cost=0.00..208.71 rows=10483 width=0) (actual
time=1.051..1.051 rows=9684 loops=1)"
"          Buffers: shared hit=39"

```

```

"                                -> Index Only Scan using
idx_lab_oi_order_cover on order_item oi (cost=0.43..3.18 rows=4 width=26) (actual
time=0.003..0.003 rows=3 loops=9684)"

"                                Index Cond: (order_id = co.order_id)"

"                                Heap Fetches: 1424"

"                                Buffers: shared hit=29663"

"                                -> Hash (cost=15.00..15.00 rows=700 width=16)
(actual time=0.444..0.445 rows=700 loops=3)"

"                                Buckets: 1024 Batches: 1 Memory Usage:
41kB"

"                                Buffers: shared hit=24"

"                                -> Seq Scan on price_list_item pli_1
(cost=0.00..15.00 rows=700 width=16) (actual time=0.273..0.355 rows=700 loops=3)"

"                                Buffers: shared hit=24"

"                                -> Hash (cost=19.00..19.00 rows=700 width=28)
(actual time=0.780..0.780 rows=700 loops=3)"

"                                Buckets: 1024 Batches: 1 Memory Usage: 52kB"

"                                Buffers: shared hit=36"

"                                -> Seq Scan on product p_3 (cost=0.00..19.00
rows=700 width=28) (actual time=0.559..0.667 rows=700 loops=3)"

"                                Buffers: shared hit=36"

"                                -> GroupAggregate (cost=84.84..95.59 rows=200 width=28) (actual
time=1.411..1.635 rows=7 loops=1)"

"                                Group Key: p_4.name"

"                                Buffers: shared hit=22"

"                                -> Unique (cost=84.84..90.09 rows=700 width=28) (actual time=1.379..1.530
rows=700 loops=1)"

"                                Buffers: shared hit=22"

"                                -> Sort (cost=84.84..86.59 rows=700 width=28) (actual time=1.377..1.440
rows=700 loops=1)"

"                                Sort Key: p_4.name, pl.bakery_id"

"                                Sort Method: quicksort Memory: 57kB"

"                                Buffers: shared hit=22"

"                                -> Hash Join (cost=33.00..51.76 rows=700 width=28) (actual
time=0.311..0.598 rows=700 loops=1)"

"                                Hash Cond: (pli_2.price_list_id = pl.price_list_id)"

```

```

"                                Buffers: shared hit=22"
"                                -> Hash Join (cost=27.75..44.60 rows=700 width=28) (actual
time=0.213..0.398 rows=700 loops=1)"
"                                Hash Cond: (pli_2.product_id = p_4.product_id)"
"                                Buffers: shared hit=20"
"                                -> Seq Scan on price_list_item pli_2 (cost=0.00..15.00
rows=700 width=16) (actual time=0.013..0.067 rows=700 loops=1)"
"                                Buffers: shared hit=8"
"                                -> Hash (cost=19.00..19.00 rows=700 width=28) (actual
time=0.188..0.188 rows=700 loops=1)"
"                                Buckets: 1024 Batches: 1 Memory Usage: 52kB"
"                                Buffers: shared hit=12"
"                                -> Seq Scan on product p_4 (cost=0.00..19.00 rows=700
width=28) (actual time=0.015..0.112 rows=700 loops=1)"
"                                Filter: is_active"
"                                Buffers: shared hit=12"
"                                -> Hash (cost=4.00..4.00 rows=100 width=16) (actual
time=0.082..0.082 rows=100 loops=1)"
"                                Buckets: 1024 Batches: 1 Memory Usage: 13kB"
"                                Buffers: shared hit=2"
"                                -> Seq Scan on price_list pl (cost=0.00..4.00 rows=100
width=16) (actual time=0.048..0.065 rows=100 loops=1)"
"                                Filter: (is_active AND (valid_from <= CURRENT_DATE)
AND ((valid_to IS NULL) OR (valid_to >= CURRENT_DATE)))"
"                                Buffers: shared hit=2"
"Planning:"
" Buffers: shared hit=69"
"Planning Time: 1.546 ms"
"Execution Time: 161.271 ms"

```

Таблица сравнения:

Запрос	Время планирования до индексов, ms	Время планирования после индексов	Время выполнения до индексов, ms	Время выполнения после индексов, ms	Процент ускорения времени выполнения
Первый	0,26	0.29	156	9.1	1714
Второй	2.2	1.55	267	161.3	66

Пояснения:

До создания индексов запросы читали крупные таблицы `customer_order` и `order_item` преимущественно через `Parallel Seq Scan`. Это означало, что просматривалось большое количество строк и затем отбрасывал неподходящие по условию `status`.

Индекс `idx_co_done_order_cover` является частичным индексом по таблице `customer_order`. Он содержит только заказы со статусами `COMPLETED` и `DELIVERED`, поэтому ускоряет оба выбранных запроса. В запросе 1 он заменяет полный просмотр `customer_order` на `Index Scan`, а в запросе 5 используется через `Bitmap Index Scan`.

Индекс `idx_oi_order_cover` является покрывающим индексом по таблице `order_item`. Его ключ `order_id` соответствует условию соединения `oi.order_id = co.order_id`, а поля `price_list_item_id`, `quantity`, `line_total_amount` добавлены через `INCLUDE`, потому что используются дальше в соединениях и агрегатах. Благодаря этому в запросе 5 появляется `Index Only Scan`, и больше не нужно просматривать всю таблицу `order_item`.

После создания индексов изменилась и `join`-стратегия. До индексов `customer_order` и `order_item` соединялись через `Parallel Hash Join`, потому что обе таблицы фактически читались большими потоками. После индексов выбирается `Nested Loop`: сначала получает подходящие заказы, затем для каждого заказа быстро находит позиции через индекс по `order_item(order_id)`.

Остальные соединения с небольшими справочными таблицами выполняются через `Hash Join`, потому что эти таблицы маленькие и соединяются по равенству.